KEVIN JOSHUA L

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Aspiring Cloud & DevOps Engineer | AWS Certified Cloud Practitioner

SUMMARY

Computer Science undergraduate building serverless and containerized systems on AWS. Five public projects cover event-driven pipelines, ML deployment, infrastructure as code, and Linux automation, with source code linked below. Seeking cloud and DevOps internships.

PROJECTS

BlockTrain: AI Railway Block Planning Digital Twin (Smart India Hackathon)

github.com/kevinjosh10/Block-Train

Next.js 14, Node.js, Python, scikit-learn, PostgreSQL (AWS RDS), Docker Compose, Amazon EC2, AWS SSM

- Built a digital twin of a 60 km Southern Railway corridor (26 stations, 72 signals, 182 switches) and an ML engine that groups maintenance tasks from track, signal, and traction departments into shared line blocks.
- On simulated data, cut weekly track closure from 116 h to 16.2 h (86% reduction) with 100% recall on the critical-defect class.
- Deployed as a multi-container Docker stack on EC2 with RDS PostgreSQL in a private VPC. Credentials are stored in SSM Parameter Store, and none are hardcoded.

CloudTrain Studio: Serverless MLOps Platform

github.com/kevinjosh10/CloudTrain

AWS Lambda, S3, SQS (DLQ), API Gateway, AWS SAM, Python 3.12, scikit-learn, GitHub Actions, React

- Built an event-driven training pipeline: an S3 upload triggers a Lambda that trains and evaluates the model, and a new version is promoted only if it beats the current one on accuracy or R^2.
- Routed failed training runs to an SQS dead-letter queue for failure isolation and inspection.
- Served predictions through API Gateway and Lambda, caching loaded models in memory across warm invocations to reduce latency on repeat requests.

Cloud-Morph: Serverless File Processing Platform

github.com/kevinjosh10/Cloud-Morph

AWS Lambda, API Gateway, Amazon S3, Python, Node.js, JavaScript, GitHub Pages

- Built a direct-to-S3 upload flow: API Gateway and Lambda issue time-limited pre-signed URLs, so the browser uploads straight to S3 and no AWS credentials ship in the frontend.
- An S3-triggered Lambda processes PDF, CSV, and text files and writes JSON insights (word counts, CSV structure) that a static dashboard renders.

Automated S3 Backup Tool

github.com/kevinjosh10/s3-backup-tool

Python, Boto3, Amazon S3

- Built an incremental backup engine that compares MD5/ETag checksums to skip unchanged files.
- Configured S3 lifecycle rules that move older backups from Standard to Infrequent Access to Glacier to reduce storage cost.

Infrastructure & Linux Projects

github.com/kevinjosh10/1000-days-ai-cloud-infrastructure-journey

Terraform, Kubernetes, Helm, Bash, systemd, auditd

- Modular 3-tier AWS architecture in Terraform (VPC, RDS Multi-AZ, ALB, Auto Scaling) with least-privilege security groups and remote state.
- Kubernetes stack with NGINX Ingress, PostgreSQL StatefulSet on EBS CSI volumes, horizontal pod autoscaling, and custom Helm charts.
- Linux monitoring suite (sysmon) with systemd timers that collects CPU, memory, disk I/O, and socket metrics with JSON export; user provisioning with Ed25519 SSH keys, scoped sudoers rules, and auditd watches.

LEADERSHIP & EXPERIENCE

Hackathon Team Lead, Smart India Hackathon (Ministry of Railways)

- Led a cross-functional team across AI, backend, and AWS deployment that pitched BlockTrain to the Ministry of Railways, with an 86% simulated reduction in track closure time.

EDUCATION

Jerusalem College of Engineering, Chennai

Expected 2029

B.E. Computer Science & Engineering, CGPA: 8.2, expected 2029

Class 12, Sri Mahalakshmi Vidyalaya (GRT), Chennai, 2025: 90% • Class 10, Alpha CBSE School, Chennai, 2023: 92%

CERTIFICATIONS & LEARNING

- AWS Certified Cloud Practitioner; AWS Skill Builder: Cloud Practitioner Learning Path, S3 Lifecycle & Storage Concepts, Cloud Storage Security Fundamentals
- 1000-Day AI Cloud Infrastructure Journey: 226+ days of public daily engineering logs and 700+ commits, with weekly roadmaps and project write-ups.

TECHNICAL SKILLS

Cloud: AWS (Lambda, S3, EC2, RDS, API Gateway, SQS, IAM, VPC, SSM, CloudWatch, SAM), Terraform, Kubernetes, Helm, Docker

DevOps & Systems: GitHub Actions CI/CD, Docker Compose, Linux (systemd, auditd, UFW), Bash, Git, Trivy, TCP/IP, DNS, VPC security

Languages: Python, TypeScript, JavaScript, Java, C, C++, SQL

Soft Skills: Willingness to learn, adaptability, self-motivation, ownership, time management, attention to detail, analytical problem-solving, troubleshooting and root-cause analysis, cross-functional communication with backend and frontend teams, collaboration with senior engineers, teamwork, customer support, technical documentation, presenting and pitching

Frameworks & ML: Next.js, React, Node.js, Express, scikit-learn, Pandas, NumPy